NUMERICAL MODELING OF COATING DELAMINATION DURING SCRATCHING

A Research Project

by

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ABSTRACT

NUMERICAL MODELING OF COATING DELAMINATION DURING SCRATCHING

(May 2019)

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The purpose of this graduate research project is to understand the delamination mechanism using the finite element analysis on ceramics and polymer coatings applied on a steel substrate. Ductile and brittle material is used with different thickness and adhesive stiffness. ASTM standard scratch testing is simulated using a three-dimensional finite element method (FEM) modeling. Delamination of coating from the substrate during scratching is studied using the FEM model. The results show that increasing the coating thickness and adhesive stiffness between the coating and the substrate increases the scratch normal load required to delaminate the coating from the substrate. This study shows that using appropriate numerical framework, the delamination failure in coatings can be modeled using FEM. The study also suggests how delamination failure can be delayed by modifying the material properties and coating thickness.

ACKNOWLEDGMENT

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CHAPTER I

INTRODUCTION

1.1 Coating Systems

The coating is an extra protective layer which is applied to protect the substrate in different scenarios. Application of coating on metal results in increase in corrosion resistance, the reduction in wear and tear. Coating protects the substrate from environmental exposure. Coatings are often made of metals, ceramics, and polymers depending on the substrate and applications. The metallic coating gives a shiny and glossy appearance. They protect the substrate from sunlight, some coating protects the substrate from oxidation and corrosion. Polymeric coating is used to increase corrosion resistance whereas ceramic coating is used to increase wear and tear resistance. Polymers are cheaper and they can easily be produced. Ceramics has high strength. Due to these advantages, industries are using coating systems to get a robust product that increases product value. The criteria of selection of the coating may be the type of substrate, environmental condition, ease of application, maintainability, mobilization, costs, appearance.

Polymer coatings are popular because it has high resistance to all type of external factors, besides this, it also provides high aesthetic value. The anticorrosion property of the polymer coating increases the life of the substrate. They are also harmless and environmentally friendly. Ceramic coatings increase the strength of the part. It also provides thermal resistance, increases hardness and wear resistance, improves durability, chemical & corrosion resistance, thermal and electrical insulation, improves thermal shock resistance. Due to these properties, nowadays, it is

used extensively used in automobile engines, headers, exhaust manifolds, turbocharger casings, downpipes, and tailpipes.

In this research project, selected coatings for investigation are TiN (Titanium Nitrate) and Acrylic. Here, TiN is a ceramic whereas Acrylic is a polymer. Use of TiN coating on steel can increase hardness, lubricity, lowers coefficient of friction and increase wear resistance [1][2]. It is used in automotive, aerospace, chemical, food processing, medical, military, and weaponry industries. Acrylic coating can mostly be applied by spraying, called acrylic spray [3]. It covers the substrate and avoids environmental exposure. As such, it increases the life of the substrate. Moreover, it is environmental friendly, can be applied easily, and is UV resistant [4].

1.2 Scratch Testing

Scratch testing is a method of characterizing surface properties of substrates or coatings. The products must be tested under different conditions to ensure that they meet the necessary requirements for the application in order to provide consumers with reliable products. Scratching on the substrate can cause failures like deformation, chipping, and cracking. Scratching on the coating-based system can cause different failures like cracking, delamination, and undercutting [5].

Delamination is a surface defect in which the surface of the part is peeled off layer by layer. It is an adhesion failure of the coating with the substrate. Factors affecting delamination are the mechanical properties of the coatings and substrates, coating thickness, type of corrosive environment, initial defects or impacts, and delamination or adhesion strength [6]. Coatings are applied on substrates to increase reliability and durability of product due to specific properties of

the coating. Scratching of the coating-based system results in inducing elastic deformation and in some case plastic deformation until delamination occurs at some critical load. Due to that, delamination is a crucial failure mode to be investigated for the reliability of the coating system. Delamination effect depends on coating thickness, coating material, and delamination or adhesion strength [7, 8]. Delamination strength is the strength of laminate's bond to a substrate material. It is also called adhesive strength or adhesive stiffness. In this research project, study of the effects of coating layer thickness, material properties and adhesive strength on coating delamination is carried out.

1.3 Literature Review

The scratch testing has been acquiring progressively more credibility for coating system. Significant researches were done on scratch resistance of various materials, however very few research works has been done in the area of coating delamination. There were many methods that have been used to conduct a study of adhesion of coatings. An early study suggested there are 17 different methods to understand the scratching mechanism [9]. ASTM standard for scratch testing is designated as a standard testing methodology.

Previous studies show different failure modes in substrate and coatings in terms of hardness of both. If soft coatings are applied on the soft substrate the material is subjected to plastic deformation and indentation development and slight or no cracking can be detected [10]. If hard coating on soft substrates is used, it results in plastic deformation in an earlier stage and then fracture of the coating can be observed [10]. If the soft coating is applied on a hard substrate, it is inclined to plastic deformation and some extrusion of the coating, which causes delamination of

the coating. For hard coating on hard substrate, scratching cause nominal plastic deformation and fracture [10]. Factors that affect the scratch test are divided into intrinsic and extrinsic. The Intrinsic factors are loading rate [11-13], scratching speed [11-13], indenter tip radius [1112], indenter wear [12], machine stiffness/design. The extrinsic factors are substrate properties (e.g., hardness, elastic modulus) [13], coating properties (e.g., thickness, hardness, modulus, residual stress) [12,13], friction coefficient [11, 14], and surface roughness [13].

Using hard coatings like TiC (Titanium Carbide), and TiN (Titanium Nitrate) on steel substrate, an increase in the wear resistance has been observed [7]. The study was conducted for TiN and TiC with different coating-substrate combinations. The critical load was termed as the load for onset of delamination. The results of the study [7] are shown in Figure 1. The study shows that increasing the thickness of the coating material increases the critical load for delamination [7].

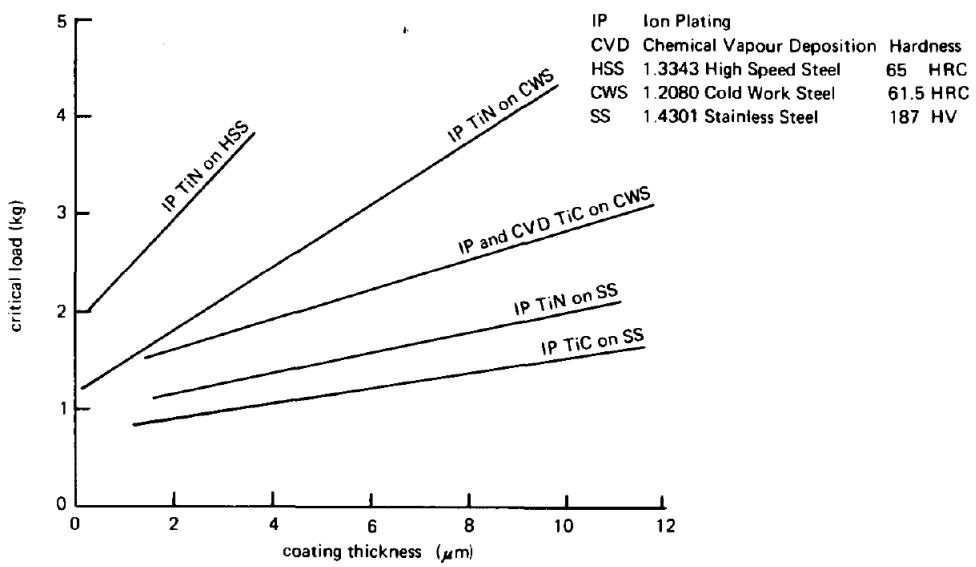


Figure 1 Critical load for delamination with respect to coating thickness for various coating-substrate combinations [7].

Another study was conducted [8] on a steel substrate with a uniform polymeric coating. In this case, the coating was soft, and the substrate was hard. Figures 2 and 3 show the images of different damage patterns observed during scratching of the coating systems. Here, Optical image of the scratch test is divided into 3 zones: Zone 1: Adhesive delamination, Zone 2: Microcracking, Zone 3: Coating failure (buckling) and exposure of the substrate.

It was concluded that moving from ductile to brittle substrate-coating composition can delay delamination of the coating in terms of critical load. Furthermore, increasing the coating thickness delays the delamination in terms of critical load [8].

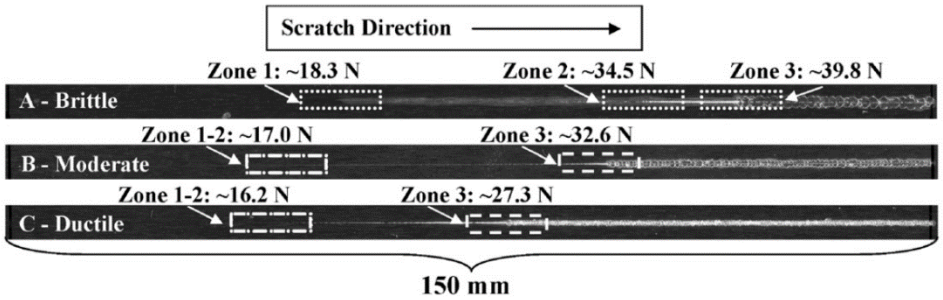


Figure 2 Scratch damages observed in different coating systems with variation in material properties [8].

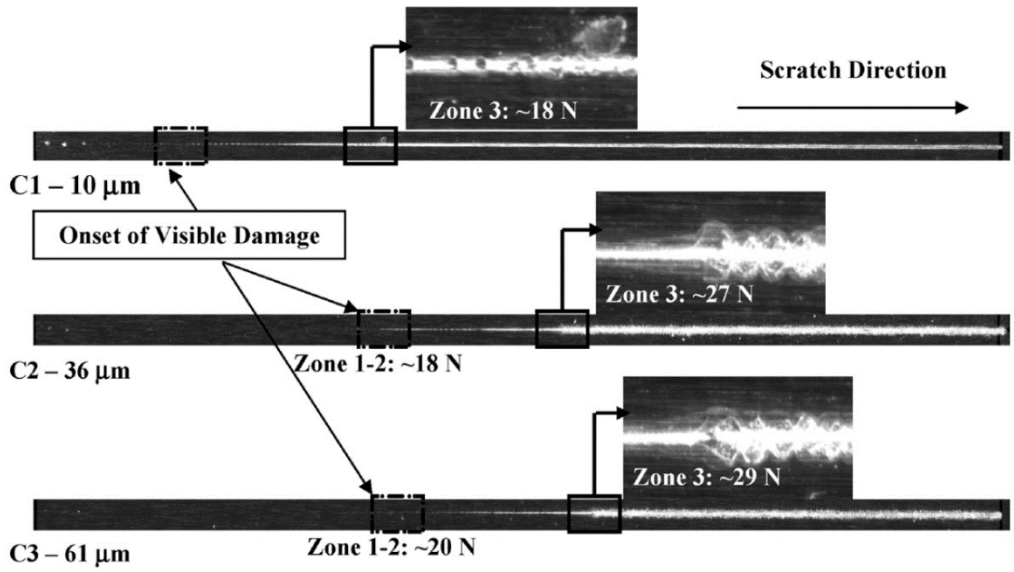


Figure 3 Scratch damages observed in different coating systems with variation in coating thickness [8].

1.4 Research Objectives

Polymeric or ceramic coatings are applied on materials to protect the surface primarily from corrosion and to increase the strength for various applications, specifically in oil and gas pipeline applications, automobile industry. Hard coatings are also applied on polymer surface to increase the surface damage resistance of a polymer. However, coating delamination is one of the primary failure mechanisms when subjected to scratching scenario.

Finite element method (FEM) modeling can facilitate understanding the delamination mechanisms during scratching and enable designing better damage resistant coatings. In this graduate research project, a three-dimensional FEM modeling is carried out to simulate the coating delamination, for ceramic and polymeric coating on steel, occurring during scratching. Specific objectives of the project are to:

1. Develop FEM models according to the ASTM standard to simulate coating delamination during scratching;
2. Analyze scratch normal load during scratching to better understand coating delamination mechanisms;
3. Study the effects of coating layer thickness, material properties and adhesive strength on coating delamination.

All the above-mentioned objectives are achieved in this project.

CHAPTER II

METHODOLOGY

To analyze the effect of thickness and adhesive strength on coating delamination, FEM (Finite Element Method) modeling is used in this study to determine critical normal load for delamination. The simulation was done using a commercial FEM package, ABAQUS 2017. Standard ASTM scratch test was simulated with the use of ABAQUS. The purpose of this research project is to study the fundamental mechanisms of coating delamination.

Figure 4 shows the FEM model of scratching, developed in this study, along with the boundary condition. To save the computation time and ease of simulation, the half model was used with certain boundary condition so that it can simulate the ASTM standard scratch test. The substrate was modeled with a dimension of 20 mm x 1.95 mm x 2 mm in 3 dimensions. The spherical Tip was made with a feature called rigid analytical body. The spherical tip is 1 mm in diameter. The coatings are applied on the substrate and made with 2 different thicknesses: 25 µm and 50 µm.

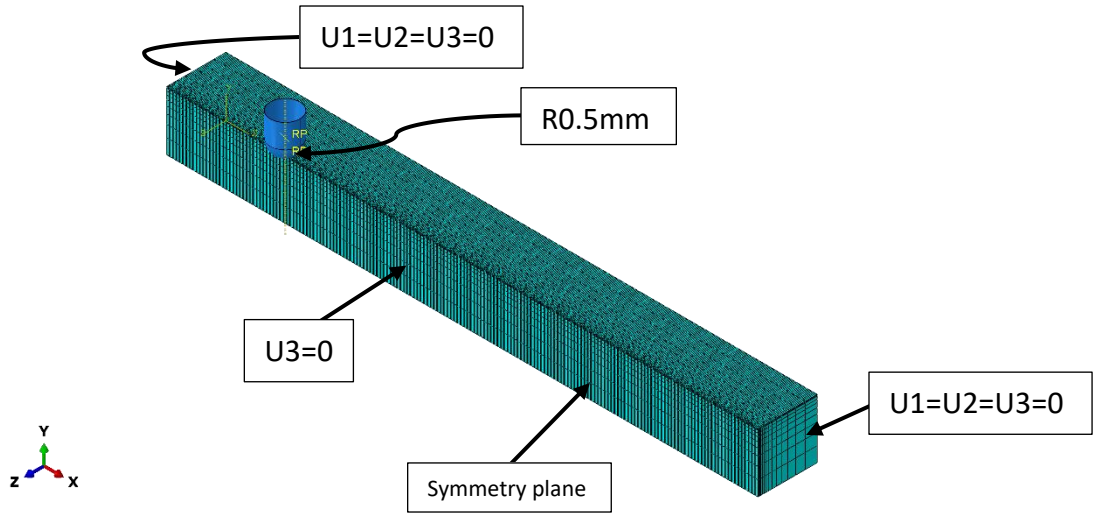


Figure 4 FEM model with meshing of scratching on substrate-coating composition

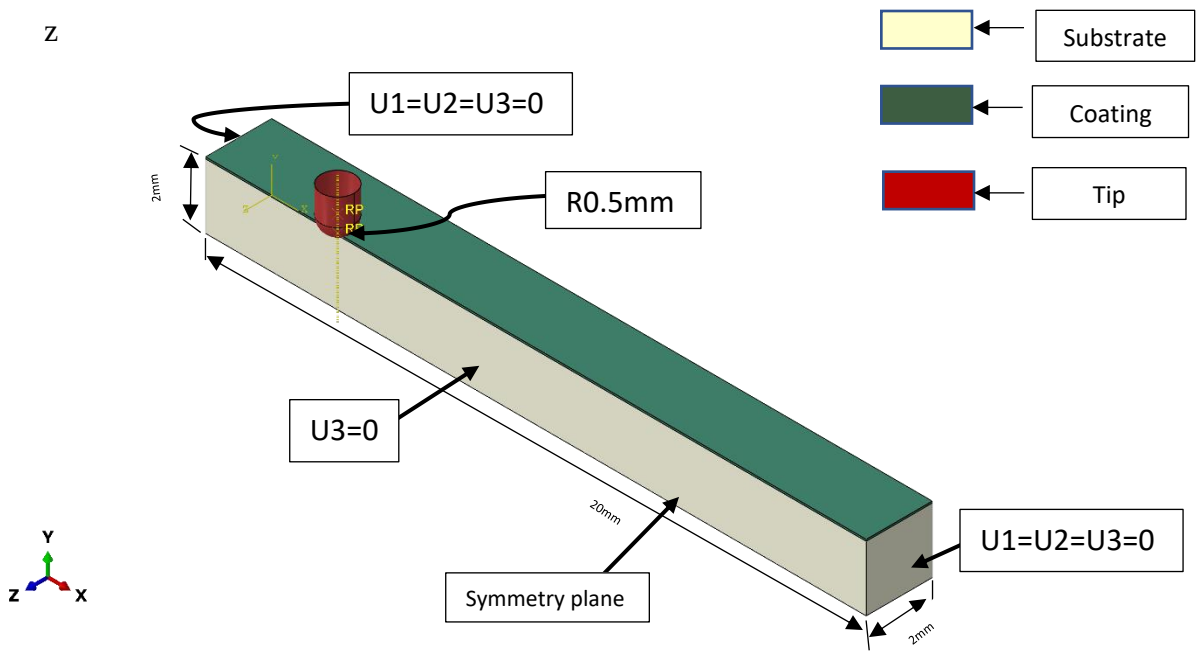


Figure 5 FEM model of scratching on substrate-coating composition.

Eight-node 3D linear brick (C3D8R) element was assigned to coating and substrate. To get a reliable and accurate result, meshing was fine in the scratching area and most of the other area was assigned biased meshing to reduce computation time. Symmetry plane was constrained in the normal direction. Node on sides and bottom was restricted to get clamping effect. Heat generation was neglected in this FEM model for scratching process. ABAQUS was used in explicit dynamic stress analysis module as scratch testing is dynamic in nature.

Simulation of scratching was divided into three steps. The first step contains indenting the substrate at a specified normal load. The second step includes the scratching of the indenter on the surface at a specified 12 mm scratch length with a certain velocity and a gradually increasing normal load of 1-30 N. In the last step tip is removed from the substrate. The FEM model was used to study the effects of coating thickness and material properties on surface damages. Coefficient of friction was used as 0.3 in all the cases and general contact was used to

define the surface interaction. Material properties of the substrate and coatings were obtained from the literature and included in the FEM model using piece-wise linear stress-strain curves. Tables 1-3 list all the material properties used in the FEM model.

Table 1 Material properties used in the FEM model for Steel [7].

Modulus of Elasticity (GPa)	Yield stress (MPa)	Ultimate strength (MPa)	Poisson's Ratio	Density (Kg/m^3)
200	215	505	0.3	8000

Table 2 Material properties used in the FEM modeling for Acrylic [8].

Modulus of Elasticity (GPa)	Yield stress (MPa)	Strain Hardening Slope (MPa)	Poisson's Ratio	Density (Kg/m^3)
3	70	100	0.4	1180

Table 3 Material properties used in the FEM modeling for TiN [7].

Modulus of Elasticity (GPa)	Ultimate strength (GPa)	Poisson's Ratio	Density (Kg/m^3)
250	1	0.25	5500

There are different methods to model delamination in ABAQUS. They are Virtual Crack Closure Technique (VCCT) and Cohesive Zone model (CZM). VCCT is used in ABAQUS/Standard to analyze brittle interfacial crack propagation. These calculations can be extremely tedious for the cases in which the crack propagation is involved [15]. CZM is a suitable candidate to model delamination. CZM has been used for simulation of delamination or separation [16,17]. It works as a traction-separation model. It is again divided into 2 method, i.e., cohesive

element method, and surface-based cohesive behavior. Surface-based cohesive behavior is used in the situation when cohesive element of very thin layer is needed. The surface-based cohesive behavior can be included in the FEM model by defining stiffness value for traction. There are 3 different direction for traction stiffness: normal direction and 2 tangential directions. They are designated as K_{nn}, K_{ss}, and K_{tt}, respectively. In this study, delamination is expected to occur due to force in the normal direction. So K_{ss} and K_{tt} will not be responsible for delamination. Therefore, a very high value of K_{ss} and K_{tt} was considered (8×10^{12} Pa) in all the cases, whereas K_{nn} value was varied (5×10^{12}Pa, 2×10^{13}Pa, 1×10^{15}Pa) to study the effect of delamination or adhesive strength on delamination.

CHAPTER III

RESULTS AND DISCUSSION

All the cases were simulated using ABAQUS 2017, using the supercomputing facility at Texas A&M University. The result on the scratch step was divided by 24 output field so that every 0.5 mm simulation observation can be carried out. The scratch length where delamination occur during scratching is obtained from the FEM simulation, and then converted into scratch normal load.

3.1 Effect of Adhesive stiffness on coating delamination

In the ABAQUS output, the 2nd step (scratch step) has 24 frames. From this, the frame was selected where the onset of coating delamination was observed. Figures 5-8 show the coating delamination observed in the FEM models with variation in different parameters. The scratch length and scratch normal load were obtained based on the steps where coating delamination occurs. All the result can be compared on the basis of scratch normal load required to delaminate the coating. For the same layer of thickness, 3 different adhesive stiffness was used for simulation.

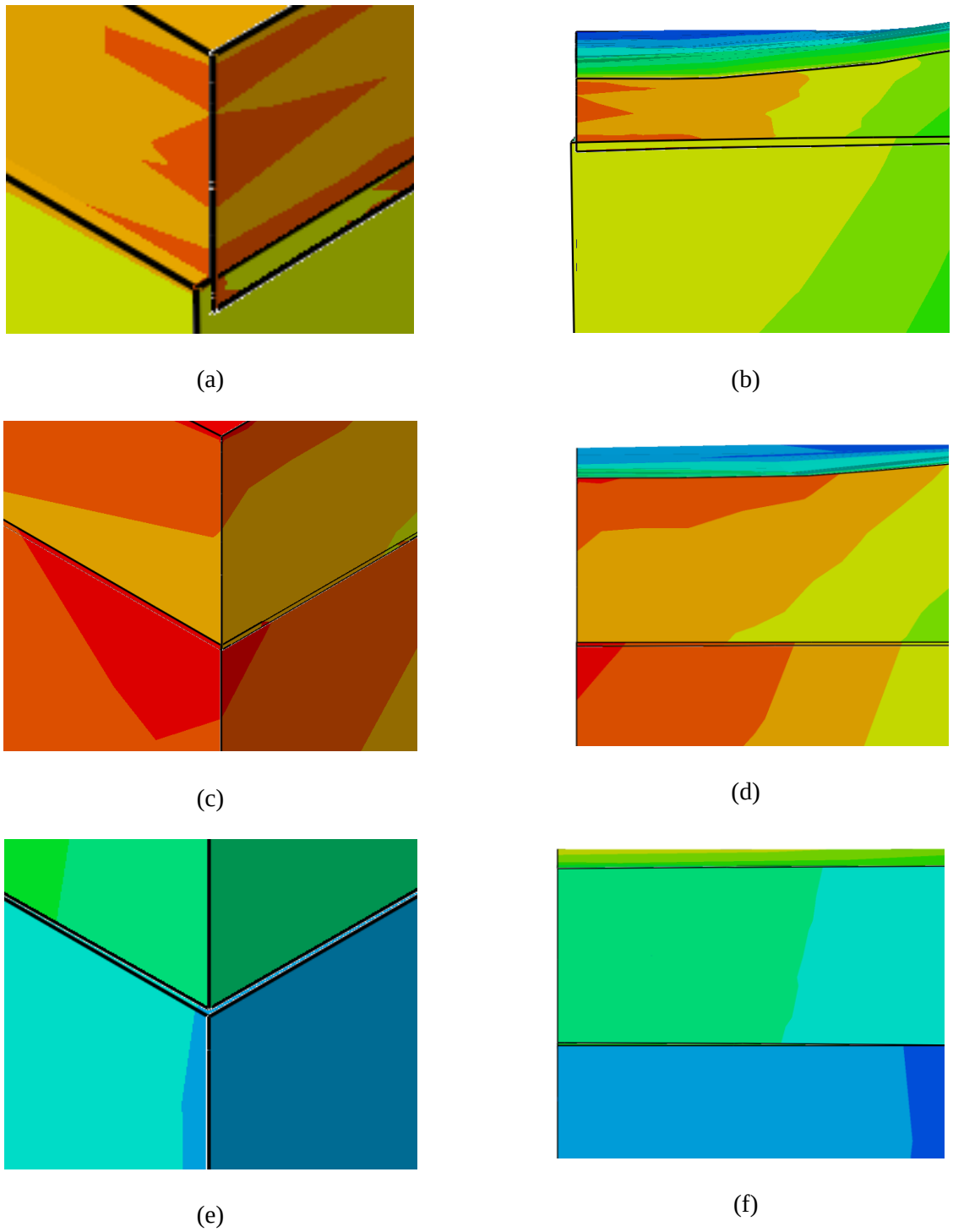


Figure 6 Delamination of 25 µm Acrylic coating (a) 3-D view, cohesive stiffness 1×10^{15}Pa (b) 2-D view, cohesive stiffness 1×10^{15} Pa (c) 3-D view, cohesive stiffness 2×10^{13}Pa (d)2-D view, cohesive stiffness 2×10^{13}Pa (e) 3-D view, cohesive stiffness 5×10^{12} Pa (f)2-D view, cohesive stiffness 5×10^{12} Pa.

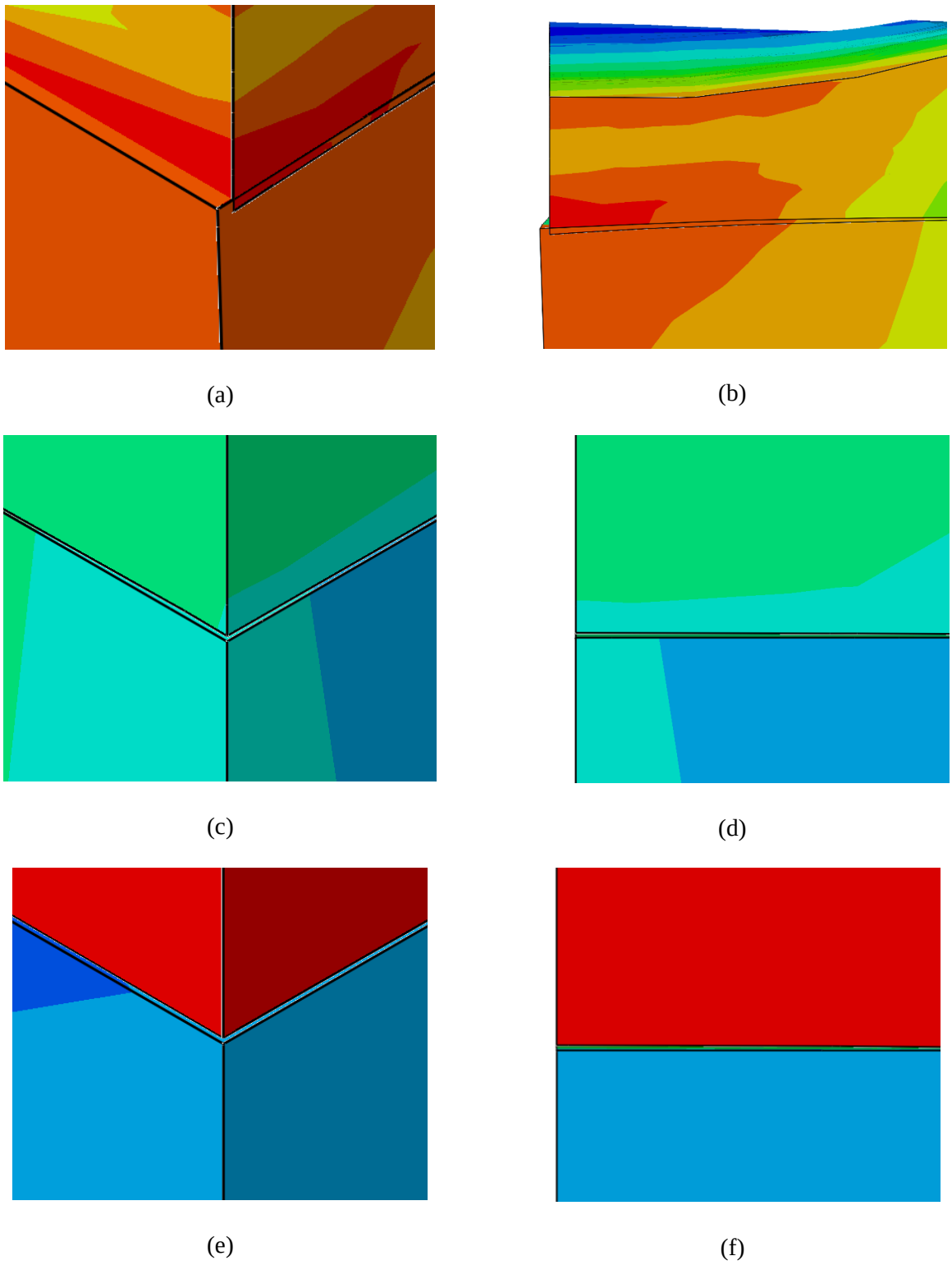


Figure 7 Delamination of 50 µm Acrylic coating (a) 3-D view, cohesive stiffness 1×10^{15}Pa (b) 2-D view, cohesive stiffness 1×10^{15} Pa (c) 3-D view, cohesive stiffness 2×10^{13}Pa (d) 2-D view, cohesive stiffness 2×10^{13}Pa (e) 3-D view, cohesive stiffness 5×10^{12} Pa (f) 2-D view, cohesive stiffness 5×10^{12} Pa.

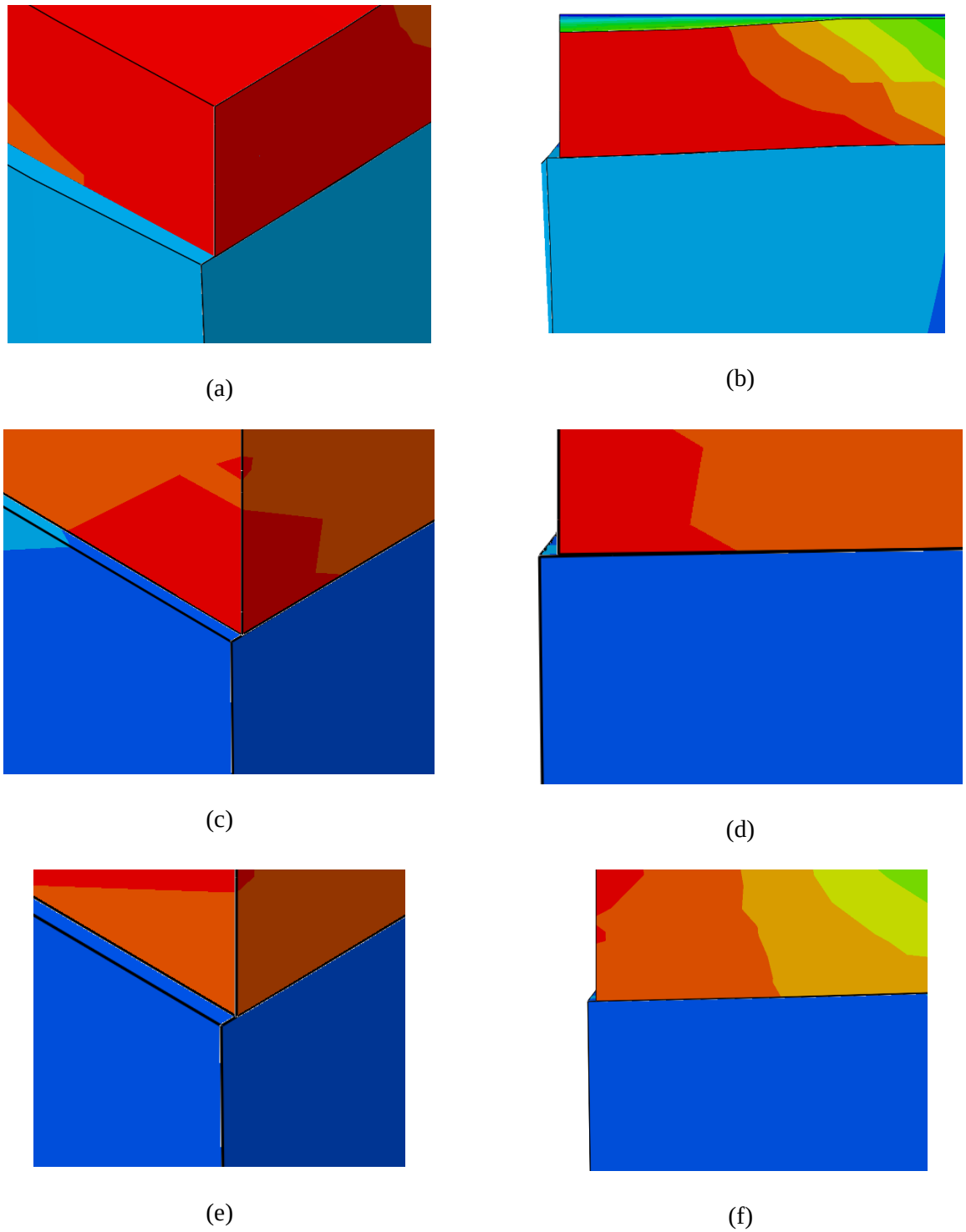


Figure 8 Delamination of 25 µm TiN coating (a) 3-D view, cohesive stiffness 1×10^{15}Pa (b) 2-D view, cohesive stiffness 1×10^{15} Pa (c) 3-D view, cohesive stiffness 2×10^{13}Pa (d) 2-D view, cohesive stiffness 2×10^{13}Pa (e) 3-D view, cohesive stiffness 5×10^{12} Pa (f) 2-D view, cohesive stiffness 5×10^{12} Pa.

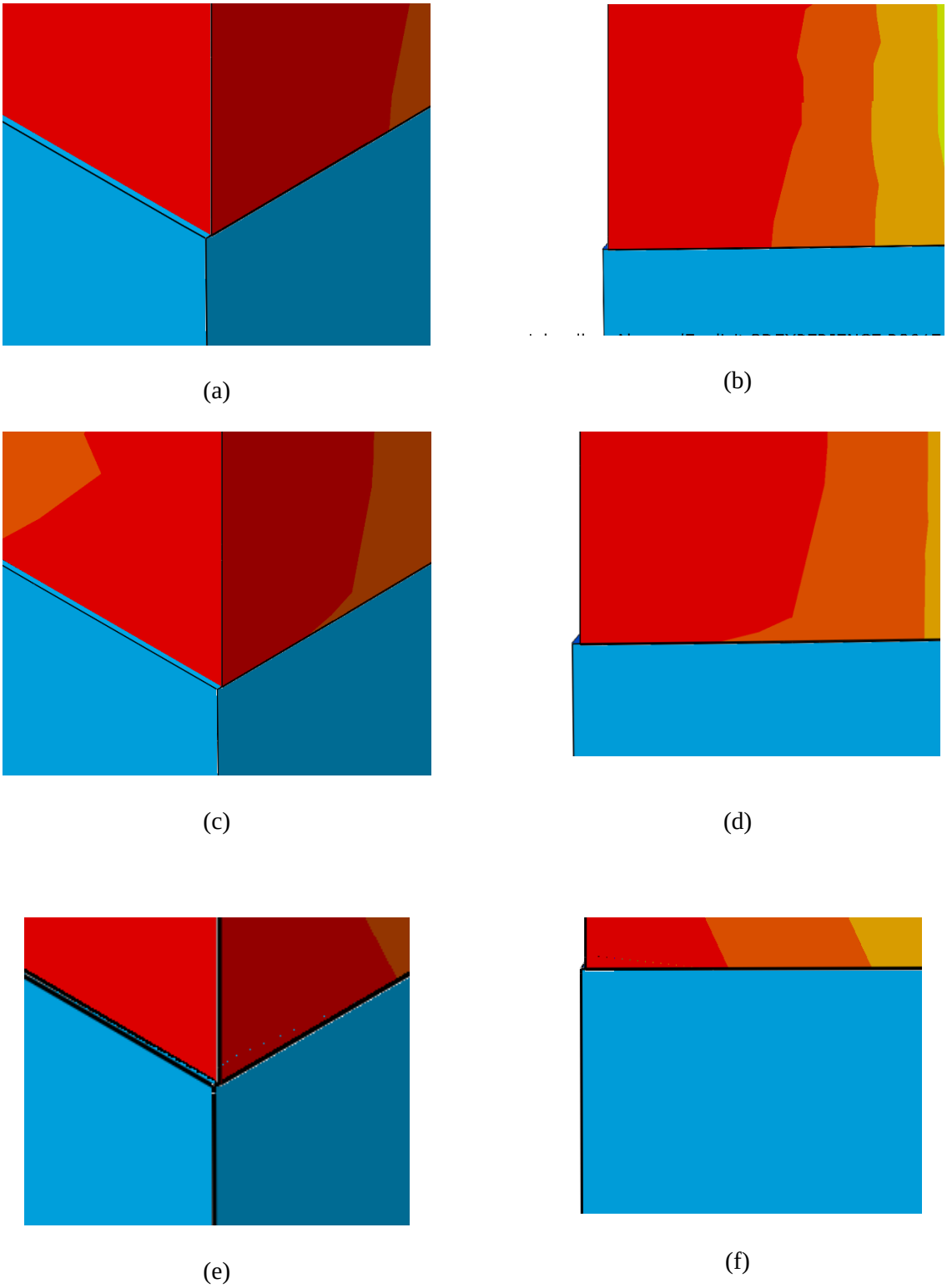


Figure 9 Delamination of 50 µm TiN coating (a) 3-D view, cohesive stiffness 1×10^{15}Pa (b) 2-D view, cohesive stiffness 1×10^{15} Pa (c) 3-D view, cohesive stiffness 2×10^{13}Pa (d) 2-D view, cohesive stiffness 2×10^{13}Pa (e) 3-D view, cohesive stiffness 5×10^{12} Pa (f) 2-D view, cohesive stiffness 5×10^{12} Pa.

Figure 9 shows the scratch normal load for coating delamination with variation of adhesive stiffness and coating thickness. It can be seen that for the increment of adhesive strength, scratch normal load required to delaminate the coating increased. For TiN coating, increasing the adhesive stiffness diminutive change in load required to delaminate the coating is observed whereas in Acrylic large change is observed.

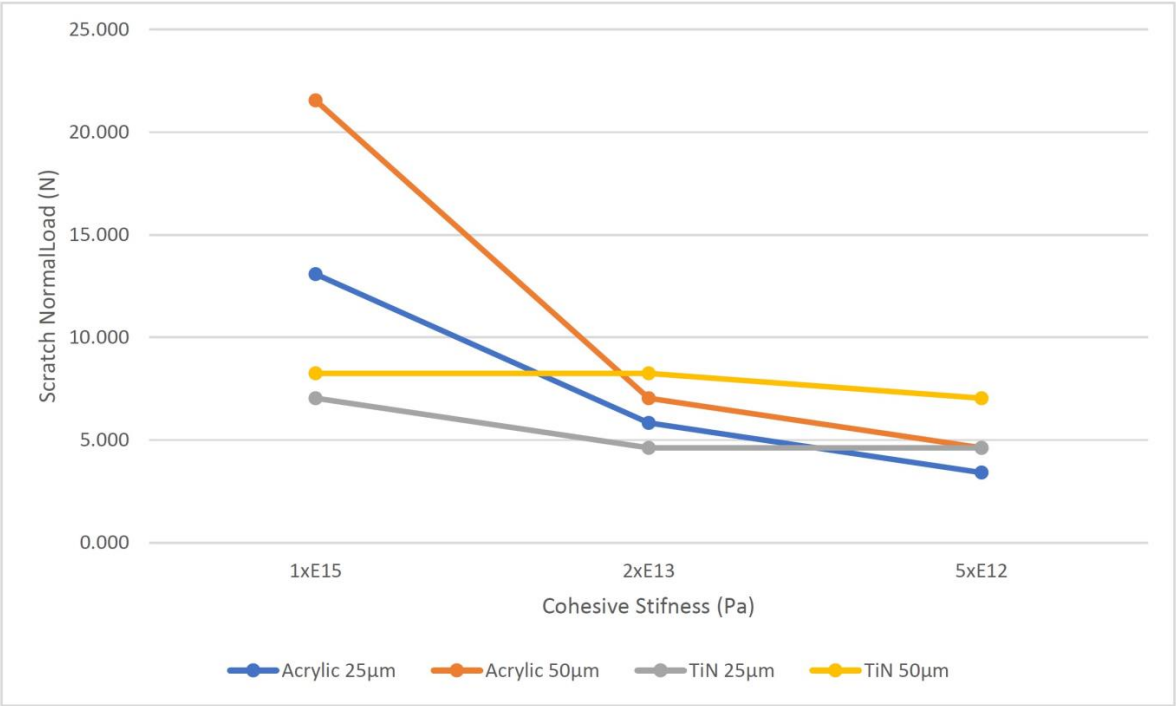


Figure 10 Scratch normal load for delamination as a function of adhesive stiffness for various coating-substrate combinations.

Figure 10 shows the scratch depth as a function of scratch length for 3 different adhesive stiffness. As can be seen in the figure, in the earlier stage during scratching, all 3 systems show similar scratch depth. This is because all 3 systems have the same material with the same coating

thickness and the same substrate. Only adhesive strength is different in these systems. As such, the scratch depth remains same for all 3 systems before delamination, which validates the FEM model.

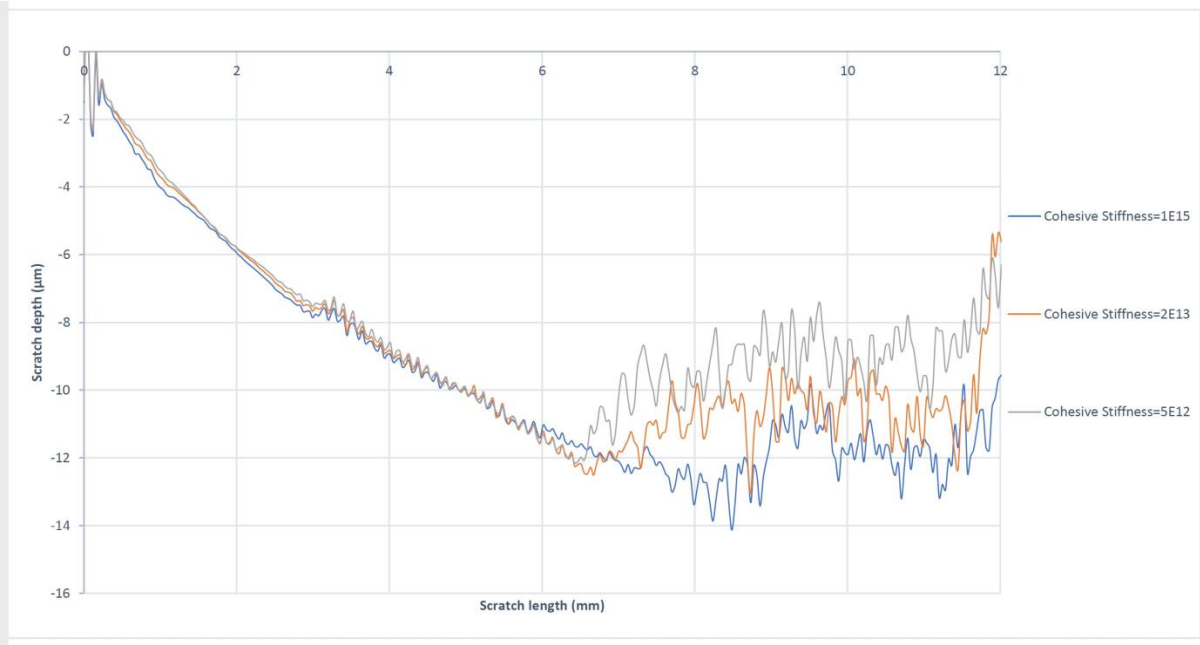
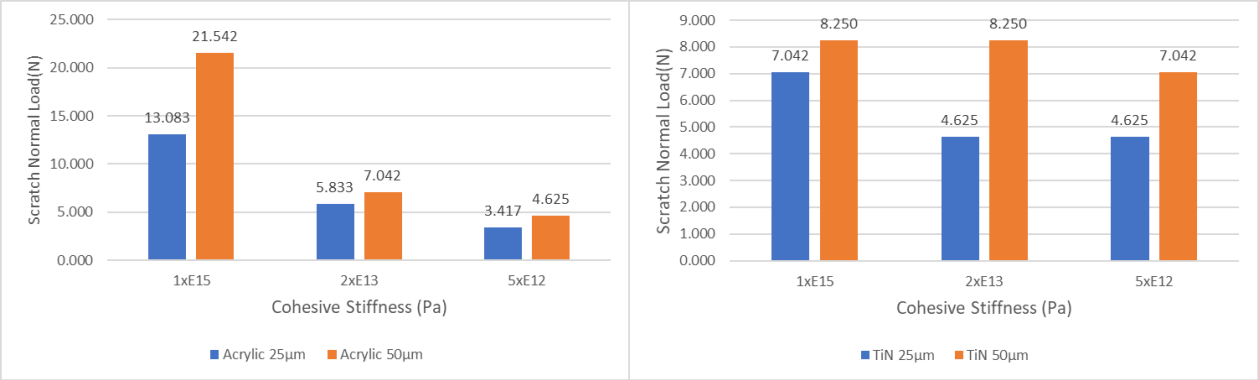


Figure 11 Comparison of scratch depth as a function of scratch length for the same coating and different adhesive stiffness.

3.2 Effect of coating thickness on delamination

Figure 11 shows the effect of coating thickness on delamination. As shown in the figure, Acrylic coating and TiN coating both show similar trend. It can be inferred from the figure that increasing coating thickness for the same adhesive stiffness increases scratch normal load required for delamination. These findings are in-line with the findings reported in the literature [7][8], which also validate the FEM model.

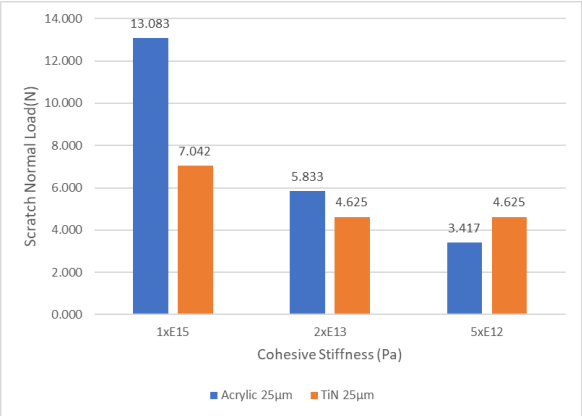


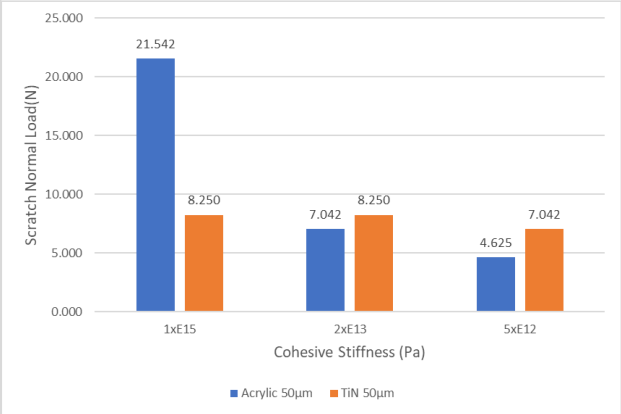
(a) (b)

Figure 12 Scratch normal load for delamination as a function of coating thickness for various coating-substrate combinations (a) Acrylic coating (b) TiN coating.

3.3 Effect of coating material on delamination

TiN is a hard and brittle ceramic and acrylic is a soft and ductile polymer. TiN can undergo only to the elastic deformation whereas acrylic can undergo elastic as well as plastic deformation. Figure 12 shows the onset of coating delamination as a function of material properties. As shown in the figure, for a higher coating thickness, TiN withstands the normal load. For other cases, ductile material has better adhesion.





(a) (b)

Figure 13 Scratch normal load for delamination as a function of coating material for various coating-substrate combinations (a) 25 µm coating (b) 50 µm.

CHAPTER IV

CONCLUSIONS AND FUTURE WORKS

Study of coating delamination mechanism is conducted in this research using FEM modeling. The FEM model was validated using the experimental results obtained from the literature. The result show that:

- Increasing the adhesive stiffness delays the onset of coating delamination;
- Increasing coating thickness increases the scratch normal load required to delaminate the coating from the substrate;
- Excluding a certain range of the coating thickness, ductile material has more adhesion compared to brittle material.

Therefore, scratch resistance of coatings applied on steel or acrylic substrate can be improved by using the results obtained in this project. The findings of this work can be used to extend the study in the following topics:

i. Investigate the failure mechanisms during scratching of coatings;
ii. Obtain optimum layer thicknesses in a multi-layer coating system to improve scratch resistance;
iii. Study the delamination failure in multi-layer polymeric films;
iv. Model the fracture behavior of single- and multilayer films.

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